

31338 Network Servers

Lab 3a: Configuring Static Networking

Aims

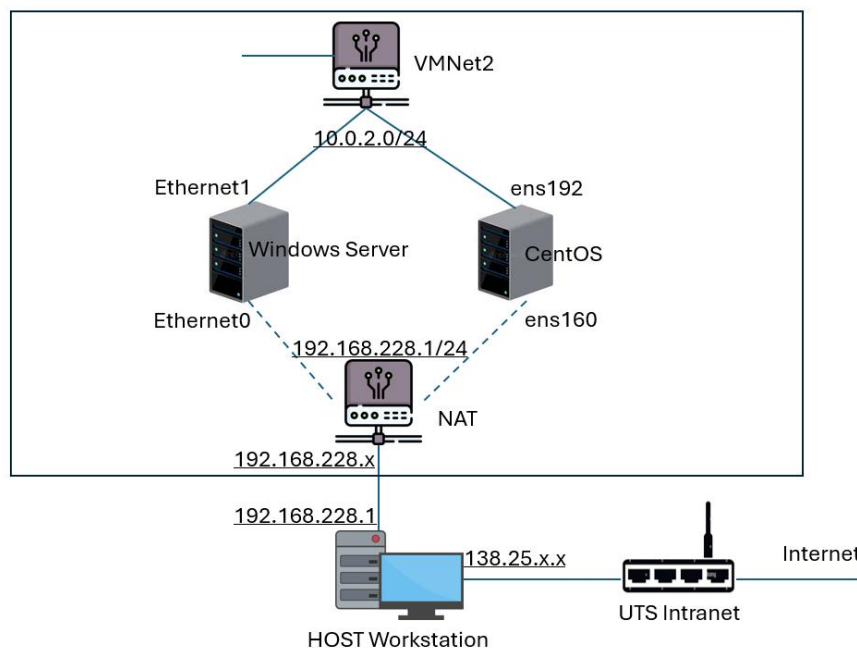
1. Use command-line tools to configure static networking
2. Make persistent network settings (for static networking) by editing networking configuration files

Task 1: Understanding the Network Topology

We assume that you are aware of the "dotted quad" conventions. This lab will use the "virtual switch" vmnet2 as the private network.

Go to *Settings* of your virtual machine to change the Network Adapter to *Custom* using *VMnet2*.

The network topology is illustrated as follows.



The configuration has the following properties.

1. The private network will be in the subnet 10.0.2.0/24
2. This lab will have to eventually cope with 200 devices
3. The gateway will be the CentOS machine
4. An appropriate IP address/netmask/gateway for both Linux and Windows Servers

Check the connection status on both Linux and Windows.

Hint: On Linux, make sure the network is up using

```
systemctl status NetworkManager.service
```

Task 2: Configuring Static Network in CentOS 10

Check the IP address setting using

```
ip addr
```

The two network adaptors ens160 and ens192 are both with DHCP by default. What are the meanings of the following properties in the terminal, *inet*, *brd* and *link/ether*? Why a valid IP address is allocated to ens160, but not ens192?

Check the default hostname by using

```
hostname
```

Change the hostname to a desired name for easier identification within the private network using

```
hostnamectl set-hostname <hostname>
```

There are several ways to setup the static network in CentOS.

1. Use the interactive tool *nmtui*;
2. Manually configure the network file */etc/NetworkManager/system-connections/<adaptor_name>.nmconnection*

It is important to understand the network configuration file. Record the settings of auto-connect, IP address, prefix, method (DHCP or manual), gateway, DNS and search domain. Also, to manage the network connection, get to know the usage of the *nmcli* command. How to shutdown/reload/activate the network connection?

View the routing table using

```
route -n
```

Notice the three columns, *Destination*, *Gateway* and *Genmask*. How do they correlate to the two network adaptors?

Finally, check the DNS server and search domain by inspecting the file */etc/resolv.conf*. Are the properties consistent with the previous settings?

Task 3: Configuring Static Network in Windows

On Windows Server, this lab will use the GUI to configure the second network adaptor, i.e., Ethernet1. In the Start menu, go to *Settings -> Network & internet -> Ethernet*, then set up the network properties.

Change the hostname in Windows (by changing the computer name).

Confirm that the configuration is correct (recall the *ipconfig* command?). What is the difference between primary DNS suffix and connection-specific DNS suffix?

Task 4: Testing & Verification

Now you should be able to ping the CentOS from Windows by using

```
ping <CentOS_IP_address>
```

from a PowerShell terminal.

Then try to ping from CentOS to Windows.

By default, Windows disables the ping command (ICMP Echo) in the firewall. To enable it, open *Windows Security -> Firewall & network protection -> Allow an app through firewall*. Check *File and Printer Sharing*.

Remember to take a snapshot of your virtual machines!